

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Antony Magrin J (192572103) (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Antony Magrin J

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.
3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS)** network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise

resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.
5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

CSA0713 – Computer Networks: Assessment Tool 3 Answers

Question 1: WLAN & WWAN Optimization – Smart Office Building

Problem Identification (Root Causes)

Issue Observed	Root Cause
Slow internet during peak hours	Too many devices (1,200+) sharing limited APs → bandwidth saturation/congestion
Frequent Wi-Fi disconnection between floors	Poor AP handoff/roaming configuration (no seamless roaming, weak overlap between AP cells)
Weak signal in conference rooms	Poor AP placement , signal attenuation through walls/floors (RF dead zones)
Video calls freezing	Insufficient bandwidth + high latency/jitter due to congestion
Some APs overloaded, others idle	Uneven load balancing — no band steering or AP load distribution
Field engineers losing WWAN in rural/highway areas	Weak/no 4G-5G cell tower coverage , cell handoff failures at high mobility
Outdated security protocols on some devices	Devices using WEP/WPA (legacy) instead of WPA3 → vulnerable to attacks

Debugging Approach

1. Conduct a **Wi-Fi site survey / heat-map analysis** to locate dead zones and overlapping/underlapping AP coverage.
2. Check **AP channel allocation** — verify no co-channel interference (adjacent APs using the same 2.4/5 GHz channel).
3. Audit **AP association logs** to confirm which APs are overloaded (client count per AP).
4. Verify **security configuration** on every AP/device via a network security scan.
5. Analyze **WWAN signal strength logs** (RSRP/RSRQ) for field engineer routes to identify coverage gaps.

Optimization Strategies

- **WLAN:**
- Deploy **additional/repositioned Access Points** for full coverage (especially conference rooms and stairwells).
- Enable **band steering & load balancing** (5 GHz for high-bandwidth devices, 2.4 GHz for IoT).
- Implement **seamless roaming (802.11r/k/v)** for smooth floor-to-floor handoff.
- Upgrade all devices to **WPA3 encryption** and disable legacy protocols.
- Use **QoS (WMM)** to prioritize video conferencing traffic over general browsing.
- **WWAN:**
- Use **dual-SIM/multi-carrier devices** for field engineers to switch to the strongest available network.
- Implement **store-and-forward/offline sync apps** so reports upload automatically when connectivity resumes.
- Consider **SD-WAN with WWAN failover** for reliable remote access.

Analysis

The core issue is a combination of **capacity mismatch (too many devices per AP)**, **poor RF planning**, and **outdated security**. Fixing AP density/placement and enabling smart load balancing directly resolves congestion and dead zones, while WPA3 migration closes the security gap — together restoring reliable, secure connectivity for all users.

Question 2: ATM Network Debugging – Financial Organization

Problem Identification (Root Causes)

Issue Observed	Root Cause
Delays in online transactions	High switch utilization , causing queuing delay for data cells
Voice calls suffer jitter/packet loss	VoIP cells possibly assigned to a wrong ATM service category (not CBR)
Video conferencing quality drops	Congested Virtual Circuits (VCs) not receiving guaranteed bandwidth
Bandwidth not efficiently allocated	Improper traffic shaping/policing and VC bandwidth reservation
Applications using wrong service category	Misconfigured QoS class mapping at the ATM switch

ATM Service Categories (Reference for Debugging)

Service Category	Best Suited For
CBR (Constant Bit Rate)	VoIP, real-time voice — needs strict guaranteed bandwidth & low jitter
rt-VBR (real-time Variable Bit Rate)	Video conferencing — needs bounded delay/jitter
nrt-VBR	Transaction processing (banking) — needs reliability, less delay-sensitive
UBR/ABR	Best-effort/bulk data transfer — no guarantees

Debugging Approach

1. Check **VC configuration logs** — confirm each application (VoIP, video, transactions) is mapped to the correct service category.
2. Monitor **switch utilization statistics** to identify congested switches/links.
3. Examine **CAC (Connection Admission Control)** settings — ensure new VCs aren't admitted beyond available capacity.
4. Verify **traffic shaping/policing parameters** (PCR, SCR, MBS) at UNI/NNI interfaces.

Optimization Strategies

- **Re-map VoIP traffic to CBR** and video conferencing to **rt-VBR** for guaranteed QoS.
- Reassign banking transactions to **nrt-VBR** (reliable, less time-critical than voice).
- Implement **traffic engineering** to redistribute load across underutilized switches/paths.
- Apply **stricter Connection Admission Control (CAC)** to prevent over-subscription of congested VCs.
- Increase **bandwidth reservation** on critical VCs carrying real-time traffic during peak

hours.

Analysis

The performance degradation stems from **QoS class mismatch and unbalanced VC/switch utilization**, not a lack of raw capacity. Correct service-category mapping plus load redistribution and stricter admission control restores the deterministic, low-jitter performance ATM is designed to provide.

Question 3: MPLS Network Debugging – E-commerce Company

Problem Identification (Root Causes)

Issue Observed	Root Cause
Slow access to cloud apps / delayed order processing	Congested LSPs carrying disproportionate traffic load
Poor voice quality, video call interruptions	Real-time traffic sharing congested paths without priority/QoS labeling
Some LSPs congested, others idle	No/poor Traffic Engineering (TE) — traffic not distributed across available paths
Incorrect label mappings on routers	Label Distribution Protocol (LDP) misconfiguration causing routing loops/inefficiency
Increased latency	Suboptimal LSP paths due to incorrect label-forwarding tables

Debugging Approach

1. Verify the **Label Forwarding Information Base (LFIB)** on each Label Switch Router (LSR) for incorrect/stale label bindings.
2. Use **MPLS traceroute/ping (LSP ping)** to confirm each LSP follows its intended path and check for label mismatches.
3. Check **LDP session status** between routers to ensure labels are being correctly advertised and learned.
4. Analyze **link utilization per LSP** to identify congested vs. underutilized paths.

Optimization Strategies

- Reconfigure and **correct label mappings** at affected LSRs (fix LDP misbindings).
- Implement **MPLS Traffic Engineering (MPLS-TE)** with **RSVP-TE** to explicitly route traffic across underutilized paths, balancing the load.
- Apply **Class of Service (CoS)/EXP-bit marking** to prioritize VoIP and video traffic over bulk data (ERP, order processing).
- Set up **Fast Reroute (FRR)** for automatic failover if an LSP fails or becomes congested.
- Regularly monitor and **rebalance LSPs** using traffic engineering tools (e.g., auto-bandwidth adjustment).

Analysis

The root problem is **absence of proper traffic engineering combined with label-mapping**

errors, causing uneven load distribution and latency. Correcting label bindings and enabling MPLS-TE with CoS prioritization ensures real-time traffic (VoIP/video) gets guaranteed low-latency paths while balancing overall backbone utilization.

Question 4: Error Control Debugging – Automated Warehouse

Problem Identification (Root Causes)

Issue Observed	Root Cause
Shipment records with incorrect info	Undetected transmission errors (weak/no error detection code in use)
Barcode scans occasionally fail	Signal attenuation/electrical interference in the wireless/wired links
Inventory mismatches in database	Corrupted frames passing through without detection , updating DB with bad data
Increased transmission errors during peak hours	Network congestion increasing collision/noise-induced bit errors
Some corrupted frames detected & retransmitted, others not	Weak/inadequate error-detection code (e.g., simple parity) — insufficient for burst/multi-bit errors

Debugging Approach

1. Identify the current error-control mechanism in use (check if it's simple parity, checksum, or CRC).
2. Analyze **frame error logs** to determine the error pattern — single-bit vs. burst errors (interference typically causes burst errors).
3. Measure **noise levels/interference sources** near scanner stations (electrical equipment, RF interference).
4. Test detection reliability of the current code against simulated burst errors.

Optimization Strategies

- **Replace weak parity checks with CRC (Cyclic Redundancy Check)**, which reliably detects burst errors — far more robust than simple parity for warehouse RF/electrical noise environments.
- For critical data (inventory counts), implement **Hamming code or Forward Error Correction (FEC)** to automatically correct single-bit errors without retransmission delay.
- Introduce **Automatic Repeat reQuest (ARQ)** protocols (e.g., Stop-and-Wait ARQ or Sliding Window ARQ) to guarantee retransmission of any frame that fails the CRC check.
- Improve physical layer: **shielded cabling, better spacing from electrical interference sources**, and stronger wireless signal design to reduce raw bit-error rate at the source.
- Add **database-level validation/checksums** as a secondary safeguard against any error that slips through the network layer.

Analysis

The core issue is that the **existing error-detection mechanism is too weak** to reliably catch burst errors caused by interference, allowing corrupted data to silently reach the database. Upgrading to CRC (detection) plus ARQ (retransmission) — or Hamming/FEC for correction —

closes this gap and ensures data integrity in the warehouse system.

Question 5: Flow Control Debugging – Online Examination Platform

Problem Identification (Root Causes)

Issue Observed	Root Cause
Incomplete downloads, delayed loading	Sender (server) transmitting faster than slow-connection students can receive
Duplicate packets, repeated retransmissions	Lost/unacknowledged packets due to receiver buffer overflow , triggering unnecessary resends
Receiver buffer overflow & app crashes	No/ineffective flow control — server sends next data block before receiver finishes processing previous one
Excessive retransmissions, increased latency	Retransmission timers firing due to overflow-induced packet loss, not actual network failure
Poor bandwidth utilization	Bandwidth wasted on redundant retransmissions instead of productive throughput

Debugging Approach

1. Monitor **TCP receive window (rwnd)** advertisements from student devices — check if the server is respecting them.
2. Analyze **packet capture logs** for retransmission rate and duplicate ACK patterns during peak login times.
3. Check whether the server implementation uses **fixed-rate transmission** instead of dynamic/adaptive flow control.
4. Identify which specific connection types (mobile vs. Wi-Fi vs. broadband) show highest error/retransmission rates.

Optimization Strategies

- Implement proper **Sliding Window Flow Control**, where the server dynamically adjusts its transmission rate based on each receiver's advertised window size (rwnd).
- Enable **TCP Congestion Control algorithms** (e.g., TCP Cubic/BBR) to adapt sending rate to actual network/receiver conditions.
- Use **adaptive bitrate/content delivery**: serve smaller, compressed question-paper files to low-bandwidth users, or use a **CDN** to distribute load geographically.
- Introduce **staggered/batched transmission** (not all 15,000 students receiving simultaneously) — e.g., randomized start-time distribution to smooth server load.
- Apply **buffer management with selective ACK (SACK)** to reduce unnecessary full retransmissions after a single lost packet.

Analysis

The fundamental problem is a **sender-receiver speed mismatch without adequate flow control**, causing buffer overflow, packet loss, and cascading retransmissions. Implementing sliding-window based flow control, congestion control algorithms, and load distribution (CDN/staggered delivery) will align the server's transmission rate with each student's actual receiving capacity, ensuring reliable delivery during peak examination load.

